

Q1.

The daily number of customers visiting a small arts and crafts shop may be modelled by a Poisson distribution with a mean of 24.

The shop offers a picture framing service. The daily number of requests, Y , for this service may be assumed to have a Poisson distribution.

Prior to the shop advertising this service in the local free newspaper, the mean value of Y was 2. Following the advertisement, the shop received a total of 17 requests for the service during a period of 5 days.

- (a) Using a Poisson distribution, carry out a test, at the 10% level of significance, to investigate the claim that the advertisement increased the mean daily number of requests for the shop's picture framing service.

(5)

- (b) Determine the critical value of Y for your test in part (a)(i).

(3)

- (c) Hence, assuming that the advertisement increased the mean value of Y to 3, determine the power of your test in part (a)(i).

(4)

(Total 12 marks)

Q2.

In a town, the total number, R , of houses sold during a week by estate agents may be modelled by a Poisson distribution with a mean of 13.

A new housing development is completed in the town. During the first week in which houses on this development are offered for sale by the developer, the estate agents sell a total of 10 houses.

- (a) Using the 10% level of significance, investigate whether the offer for sale of houses by the developer has resulted in a reduction in the mean value of R .

(6)

- (b) Determine, for your test in part (a), the critical region for R .

(2)

- (c) Assuming that the offer for sale of houses on the new housing development has reduced the mean value of R to 6.5, determine, for a test at the 10% level of significance, the probability of a Type II error.

(4)

(Total 12 marks)

Q3.

The committee of a Former Pupils' Association claims that the mean age, μ , of the association's members is more than 48 years. The standard deviation of the ages of the association's members is known to be 8 years.

To investigate the committee's claim, a random sample of 100 members is taken and the mean age is found to be 49.2 years.

- (a) Stating null and alternative hypotheses and using the 5% level of significance, test the committee's claim.

(7)

- (b) Calculate the probability of a Type II error when $\mu = 50$ years.

(5)

(Total 12 marks)

Q4.

The marks, X , obtained by students in their mathematics examination may be assumed to be a random variable with mean μ and variance 32.

After the introduction of a new programme of revision classes, it is claimed that the value of the mean has increased from its previous value of 65.

In order to investigate this claim, a random sample of 50 students' marks is taken and the mean, $\bar{x}$, calculated.

- (a) Write down null and alternative hypotheses in order to investigate the claim.

(2)

- (b) Assuming H_0 to be true, write down the distribution of $\bar{X}$.

(2)

- (c) It is decided to reject the null hypothesis if the sample mean exceeds 66.4.

Calculate the probability of a Type I error and hence write down the significance level of this test.

(4)

- (d) Calculate the probability of a Type II error in the case when μ has increased to 67.

(4)

(Total 12 marks)

Q5.

A golf club sells bottles of water in its professional's shop. The amount of water, Y millilitres, in a bottle has a normal distribution with a mean of μ and a standard deviation of 4.

The supplier claims that the mean amount of water per bottle is 300 ml. The golf club members suspected that, over a period of time, the mean amount of water per bottle had decreased. In order to investigate their suspicion, they decided to take a random sample of 20 bottles. From this sample, their calculated value for $\bar{Y}$, the mean amount of water per bottle, is 298.1 ml.

- (a) Stating null and alternative hypotheses, test, at the 1% level of significance, their suspicion that the mean amount of water per bottle is now less than 300 ml.

(6)

- (b) For the 1% level of significance, calculate the critical region for $\bar{Y}$, giving your answer to one decimal place.

(2)

- (c) Hence calculate the probability of a Type II error given that in fact $\mu = 296.5$.

(3)

(Total 11 marks)

Q6.

The volume, Y millilitres, of washing-up liquid contained in a plastic bottle is a normal random variable with mean μ and standard deviation 8.

It is suspected that μ has a value which is less than the specified value of 500.

To test this suspicion, a random sample of 100 bottles of washing-up liquid is selected and the mean volume, $\bar{Y}$, of liquid per bottle is calculated.

- (a) Write down null and alternative hypotheses for this test.

(1)

- (b) State the distribution of $\bar{Y}$, assuming the null hypothesis to be true.

(2)

- (c) Assuming a 5% significance level:

- (i) write down the probability of a Type I error;

(1)

- (ii) determine the critical region for $\bar{Y}$, giving its limit to two decimal places.

(3)

- (d) Hence determine the probability of a Type II error when $\mu = 497.5$.

(7)

(Total 14 marks)

Q7.

A random variable X is normally distributed with mean μ and standard deviation 0.8. The null hypothesis $H_0 : \mu = 40$ is to be tested against the alternative hypothesis $H_1 : \mu \neq 40$ using the 5% level of significance.

The mean, $\bar{X}$, of a random sample of 50 observations of X is to be used as the test statistic.

- (a) Write down:

- (i) the distribution of $\bar{X}$ assuming H_0 is true;

(3)

- (ii) the probability of a Type I error.

(1)

- (b) Calculate the acceptance region for $\bar{X}$, giving its limits to two decimal places. (5)
- (c) (i) Explain what is meant by a Type II error. (1)
- (ii) Calculate the probability of a Type II error in the case when $\mu = 40.3$. (5)
- (Total 15 marks)

Q8.

Santos, a newsagent, employs schoolchildren to deliver newspapers. Marilyn is to start delivering morning newspapers. She wishes to complete her delivery round early enough to ensure that she will not be late for school. Santos tells her that the average time taken to complete a delivery round is 40 minutes. Marilyn's friend Lula, who already delivers morning papers for Santos, says that the average time taken to complete a delivery round is 50 minutes.

Before starting her employment, Marilyn checks the times taken by 8 schoolchildren, employed by Santos, to complete their rounds. The times, in minutes, may be regarded as a random sample from a normal distribution and were as follows:

52 30 41 49 61 33 52 54

- (a) (i) By carrying out a hypothesis test, using the 5% significance level, show that the null hypothesis that the mean time to complete a round is equal to 40 minutes is accepted. (12)
- (ii) By carrying out a hypothesis test, using the 5% significance level, show that the null hypothesis that the mean time to complete a round is equal to 50 minutes is accepted.
- (b) The following three claims refer to the two tests carried out in part (a).

Claim 1 A Type I error has occurred in at least one of the tests.

Claim 2 A Type II error has occurred in both tests.

Claim 3 A Type II error has occurred in at least one of the tests.

For **each** of the three claims, state whether it is:

- A** definitely true;
- B** possibly true;
- C** definitely not true.

Explain your answers.

(6)
(Total 18 marks)

Q9.

A make of car battery is advertised as having a mean lifetime of at least 40 months. A car magazine suspects that the mean lifetime is less than 40 months and so monitors the lifetimes, in months, of a random sample of 12 batteries, with the following results:

44 32 46 39 38 30 35 40 38 41 34 36

- (a) Assuming that lifetimes are normally distributed, investigate the magazine's suspicion, using the 5% significance level.
- (b) A larger random sample of 160 car batteries is now monitored. The lifetimes are found to have a mean of 39.2 months and a standard deviation of 4.2 months. Use this information to investigate the magazine's suspicion, using the 5% significance level.
- (c) State, with an explanation, which, if either, of the tests in parts (a) and (b) has the larger risk of:
- (i) a Type I error;
 - (ii) a Type II error if the mean lifetime is 40 months;
 - (iii) a Type II error if the mean lifetime is 39 months.

(10)

(4)

(6)

(Total 20 marks)